Test Sheet

The 2026 Raleigh Division B Invitational, Presented by NC BioNetwork

IBM Circuit Lab

DO NOT WRITE ON THIS TEST
Open this booklet when the event leader starts time
Tiebreakers: [40, 24, 18, 12, 6, 36]

Matching (1 pt. per question)

Scientist	Contribution
1. Ampere	A. Discovered the equation describing the magnetic field created by two current-carrying wires
2. Coulomb	B. Invented wireless telegraphs through electromagnetic waves
3. Faraday	C. Measured small charges with torsion balance
4. Hertz	D. Invented the first AC electric motor and a fully remote controlled boat
5. Kirchoff	E. Invented the battery in 1800
6. Ohm	H. Invented the electric motor in 1821
7. Tesla	G. Discovered the current from a battery is directly proportional to its voltage, and inversely proportional to the resistance of the wire
8. Volta	F. Discovered the conservation of voltage in a circuit and conservation of current at a node

Calculations (2 pts per question):

9. There are 4 AA batteries with a voltage of 1.5V, each connected in series. If one battery is removed from the line, what would a voltmeter read from the new string of batteries?

10. A transformer with a rating of 500 mega volt-amps connects a current cable from a 50kV generator to a cable line that can handle 10,000 Volts. What is the turn ratio of the transformer?

11. A motor is rated for 4800 watts at 600 volts. This motor is then connected to a 240 volt outlet. What is the power (in watts) consumed by the motor?

12. A wire has a resistance of 2.4 ohms per 1000 feet. How much resistance (in ohms) will a 50 foot piece of this wire have?

Multiple Choice (1 pt. per question):

13. The volt is the unit to describe which of the following?
    a. The flow of electrons from one point to another
    b. The electric potential difference between two points
    c. The amount of power consumed by a circuit element
    d. The amount of energy a capacitor can hold

14. Which of the following units is used for current?
    a. Amp
    b. Farad
    c. Volt
    d. Watt

15. A charged particle has an electric field with arrows pointing away from the particle. What kind of particle is it?
    a. Electron
    b. Neutron
    c. Neutrino
    d. Proton

16. A standard inverter does which of the following?
    a. Converts alternating current from one frequency to another
    b. Converts a direct current to an alternating current
    c. Converts an alternating current to a direct current
    d. Converts a low voltage with a high current to a higher voltage with lower current

17. Which of the following is not a potential result of an electrical shock?
    a. Spiky hair
    b. Burns
    c. Bleeding
    d. Unconsciousness

18. A capacitor is composed of two metal plates and a _____ in between them?
    a. Conductor
    b. Dielectric
    c. Ferromagnet
    d. Inductor

19. If three 300 ohm resistors are connected in parallel, the equivalent resistance would be which of the following:
    a. 100 Ohms
    b. 900 Ohms
    c. 300 Ohms
    d. 27000000 Ohms

20. What is the purpose of connecting batteries in series
    a. To decrease the drop voltage
    b. To increase the current
    c. To increase the voltage
    d. To decrease the voltage

21. The resistance of a wire is inversely proportional to which of the following:
    a. Length
    b. Temperature
    c. Young's Modulus
    d. Cross Sectional Area

22. A relay does which of the following:
    a. Melts in the event of too much current, breaking the circuit and protecting other components
    b. A mechanical mean to open and/or close a circuit when given a specified electrical input
    c. Takes one voltage and converts it into a different voltage so a circuit can better handle it
    d. A mechanical mean to open and/or close a circuit when given a specified human input

23. An LED is put in series with a resistor and a voltage source. The resistor is replaced with a wire. What is going to happen?
    a. The light output by the LED will increase
    b. The light output by the LED will decrease
    c. The light output by the LED will not change
    d. The light output by the LED will be nothing

24. A perfectly vertical wire is sending electrons downwards. In what direction does the magnetic field induced by this charge movement go?
    a. With the current
    b. Counterclockwise
    c. Clockwise
    d. Against the current

25. How would one go about increasing the capacitance of a capacitor?
    a. Change the plate material
    b. Increase the distance between the plates
    c. Increase the plates' cross sectional area
    d. Increase the voltage across the plates

True or False (1 pt. per question):

26. Direct current is more dangerous than alternating current.

27. Resistance is directly proportional to the material's temperature.

28. A line of electrical force pointing into a charged object indicates that the charge is positive.

29. LED stands for Light Electropotential Differentiator.

30. Power is voltage multiplied by current.

31. Power factor is for AC circuits only.

32. A megohmmeter can be used to measure capacitance.

33. A rectifier converts AC to DC.

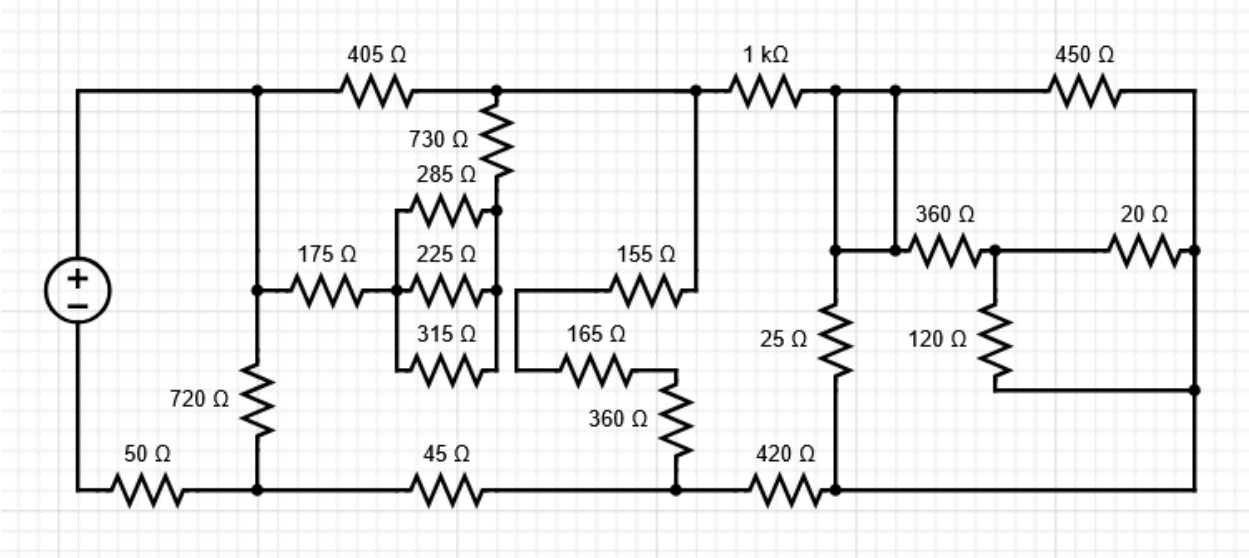
34. A resistor is put in parallel with a voltage source. The resistor is switched out for a wire. The power consumed by the wire is more than what was consumed by the resistor.

35. The product of the current and voltage of one side of a transformer is equal to the product of the current and voltage of the other side.

36. Magnetic poles ONLY come in pairs.

37. A breaker is a one-time use protection device.

Circuit Analysis (49 pts)



38. Above is a circuit diagram. Determine the equivalent resistance of the network rounded to the nearest ohm. (13 pts)
39. The current exiting the voltage source of the circuit from question 38. (12 pts)
40. The voltage across the 225 ohm resistor. (13 pts)
41. The power consumed by all of the resistors together from questions 38, 39, and 40. (11 pts)

Lab Configuration (10 pts)

Imagine you have three 100 Ohm resistors and four 200 Ohm resistors. Arrange them in such a manner that the equivalent resistance is 50 Ohms. Draw out your equivalent resistance network.